

"MTH 644"

Current Paper: 2023

MCG's

→ hamisha ring ko belong krta h.

- 1) For non-empty set class $\{\emptyset, A\}$ is
i) σ -Ring ii) Field iii) σ iv) vector

- 2) A ring on set X also Algebra Leet#
i) True ii) False

- 3) Let X be a set A is sigma on X if
 $A_1, A_2 \in A$
i) $A_1 \cap A_2 \notin A$ ii) $A_1 \cap A_2 \in A$

- 4) $I = (0, 1) \Rightarrow \inf(I) =$ _____
i) 0 ii) 1 iii) ∞

- 5) $\mu(A \Delta B) =$ _____ ?
a) $\mu(A-B) - \mu(A+B)$ b) $\mu(A-B) + \mu(B-A)$

- 6) if A is countable set then
a) $m^*(A) \geq 0$ b) $m^*(A) = 0$
c) $m^*(A) = \infty$ d) $m^*(A) \leq \infty$

- 7) if $\{I_n\}$ be a finite set of inverse intervals covering the rational in $[0, 1]$, then
a) $\sum l(I_n) \geq 1$ b) $\sum l(I_n) = 1$
c) $\sum l(I_n) \leq 1$

8) A & B are two sets with $m^*(A) = 0$ then

a) $m^*(A) \leq m^*(B)$ b) $m^*(A \cup B) = m^*(B)$

c) $m^*(A) = m^*(B)$ d) $m^*(B) \leq m^*(A)$

9) Let $\mathcal{A}$ be σ -Algebra on X if A_1, A_2 Then

a) $A_1 \cap A_2 \in \mathcal{A}$ b) $A_1 \cap A_2 \notin \mathcal{A}$

10) $\mu(A \Delta B) = \underline{\hspace{2cm}}?$

a) $\mu(A-B) - \mu(A+B)$ b) $\mu(A-B) + \mu(B-A)$

11) if $(a, b) \in U^m \bar{I}_n$ then the following true.

a) $m^*(a, b) \geq m^*(\bigcup_{n=1}^{\infty} \bar{I}_n)$

b) $m^*(a, b) \leq m^*(\bigcup_{n=1}^{\infty} \bar{I}_n)$

12) the measure $\mu(\bigcup_{n=1}^{\infty} A_n) = 0$ then obviously

a) $\mu(A_n) = 0$

b) $\mu(A_n) \geq 0$

c) $\mu(A_n) = 1$

d) None of these

13) if $\phi \neq A$; $B \in \mathcal{A}$ such that $B \subseteq A$ then

a) $\mu(B) \leq \mu(A)$

b) $\mu(A) \geq \mu(B)$

c) $\mu(A) = \mu(B)$

14) if $X \neq \phi$ & $\mathcal{A}$ be any class of subsets then $A' \in \mathcal{A}$ is same as

a) $X - A \in \mathcal{A}$

b) $\phi - A \in \mathcal{A}$

c) $X \cap A \in \mathcal{A}$

15) Every σ -ring is $\underline{\hspace{2cm}}?$

a) ring

b) σ -Algebra

16) Every ring is σ -algebra?
a) True b) False

17) For any set A & any $\epsilon > 0$, there exist an open set O containing A such that

a) $m^*(O) > m^*(A) - \epsilon$

b) $m^*(O) \leq m^*(A) - \epsilon$

c) $m^*(O) \geq m^*(A) + \epsilon$

d) $m^*(O) \leq m^*(A) + \epsilon$

Handouts

18) A is Not a σ -ring.
 $\bigcup_{i=1}^{\infty} \{n_i\} \notin \mathcal{A}$

19) $X \neq \emptyset$, $P(X) = \{ \underline{A} \subseteq X \}$

18) Every algebra is a Ring.

19) For a $X \neq \emptyset$, The ring $\mathcal{A}$ of subset of " X " is also an Algebra if & only if $X \in \mathcal{A}$.

20) A is said to be σ -Algebra if
 $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$

21) For $X = \mathbb{N}$, $\mathcal{A} = \{ A \subseteq \mathbb{N}; |A| < \infty \vee A^c \text{ is Countable} \vee A^c \text{ Countable} \}$.

22) if $X \in \mathcal{R}$, $\mathcal{A} = \{ A \subseteq \mathbb{R}; |A| < \infty \vee A^c \text{ Countable} \}$
 $\mathbb{Z} \subseteq \mathbb{R}$ but $\mathbb{Z}^c$ is not Countable.

23) σ -ring $\mathcal{A}$ is also a σ -algebra if
& only if $X \in \mathcal{A}$.

24) $A \in \mathcal{A}_\alpha \forall \alpha \in I$ But each $\mathcal{A}_\alpha$ is σ -algebra.

25) Let $\{A_i\}_{i=1}^\infty$ be a sequence of subsets if $X \neq \emptyset$,
such that $\{A_i\}_{i=1}^\infty \subseteq \bigcap_{\alpha \in I} \mathcal{A}_\alpha \Rightarrow$
 $\{A_i\}_{i=1}^\infty \subseteq \mathcal{A}_\alpha \forall \alpha \in I$.

26) $\mathcal{A}, \cup \mathcal{A}_i$ is not a σ -algebra.

27) For $X \neq \emptyset, A \subseteq X$ & $B = \{A\}$, Then $\{\emptyset, X, A, A^c\}$
is a σ -algebra generated by 'B'.

28) $\bigcap_{i \in I} \{A_i : A_i \text{ is a } \sigma\text{-algebra}, B \subseteq A_i \forall i \in I\}$

29) Arbitrary intersection of σ -algebra is
a σ -algebra.

30) $F_n \cap F_m = \emptyset, \forall m \neq n$

31) $\bigcup_{i=1}^\infty E_i = \bigcup_{i=1}^\infty F_i$

32) suppose that for any $\varepsilon > 0$, if
 $a < b + \varepsilon$, then $a \leq b$.

33) Let $X \neq \emptyset, \mathcal{A} \subseteq P(X)$, be a class
of subset of X , then a function defined

on X as:

$$\mu: A \rightarrow [0, \infty) \cup \{\infty\}$$

$$34) [0, \infty) \subseteq \mathbb{R} \Rightarrow \mu[0, \infty) = \infty$$

$$(0, 1) \subseteq \mathbb{R} \Rightarrow \mu(0, 1) = 1 - 0 = 1$$

$$35) \text{ Finitely subadditive if } \mu\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n \mu(E_i)$$

$$36) \text{ Countably subadditive if } \mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i)$$

$$37) \text{ Finitely additive if } \mu\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n \mu(E_i) \forall E_i \cap E_j = \emptyset \forall i \neq j$$

$$38) \text{ Countably Additive if } \mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i) \forall E_i \cap E_j = \emptyset \forall i \neq j$$

$$39) \text{ Monotone if } \mu(A) \leq \mu(B) \text{ whenever } A \subseteq B$$

$$40) \mu \text{ is Countably additive: } \mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i) \forall E_i \cap E_j = \emptyset, i \neq j$$

$$41) \mu(B) < \infty, \text{ then } \mu(A) - \mu(B) = \mu(A-B)$$

$$42) \text{ if } \emptyset \neq A, B \in \mathcal{A} \text{ such that } B \subseteq A, \text{ then } \mu(B) \leq \mu(A)$$

43) For $X \neq \emptyset$, let $\mathcal{A}$ be the σ -algebra on X . Then the triple $(X, \mathcal{A}, \mu)$ is called measurable space & members of $\mathcal{A}$ are called measurable sets.

44) Let $(X, \mathcal{A}, \mu)$ be measurable space then if $\exists$ a $\emptyset \neq A \in \mathcal{A}$ such that $\mu(A) < \infty$; then $\mu(\emptyset) = 0$.

$$45) \mu(A) = \mu(A \cup \emptyset) = \mu(A) + \mu(\emptyset)$$

46) Let $\{A_n\}$ is a sequence of pairwise disjoint subsets of X is σ -algebra $\mathcal{A}$ then for any $T \in \mathcal{A}$; $\mu(T) =$

$$\sum_{n=1}^{\infty} \mu(A_n \cap T) + \mu(T \cap (\bigcup_{n=1}^{\infty} A_n)^c)$$

47) All $T \cap A_n$'s are disjoint by additive property of μ .

48) In a measure space $(X, \mathcal{A}, \mu)$, if $\mu(E \cap F) = 0$, then $\mu(E) = \mu(F) = \mu(E \cup F) = \mu(E \cap F)$.

49) For $\emptyset \neq X$, the triple $(X, \mathcal{A}, \mu)$ is said to be finite measure space if $\mu(X) < \infty$, & μ is called Finite Measure.

50) For $\phi \neq X$, the triple (X, A, μ) is σ -algebra
 finite measure if & only if $\mu(X) < \infty$
 & $\exists$ a sequence $\{A_n\}$ of subsets of X
 such that

$$\bigcup_{n=1}^{\infty} A_n = X \text{ \& \; } \mu(A_n) < \infty \quad \forall A_n \in A$$

51) For any seq: $\{A_n\}$ in A of disjoint pairs

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n), \quad \nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \nu(A_n)$$

52) if $\{A_n\}$ be a sequence of subset of
 "X" i.e. $\{A_n\}_{n=1}^{\infty} \subseteq P(X)$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

53) For any arbitrary seq: $\{A_n\}_{n=1}^{\infty} \subseteq P(\mathbb{R})$ $m^*\left(\bigcup_{n=1}^{\infty} A_n\right)$

$$\leq \sum_{n=1}^{\infty} m^*(A_n)$$
 i.e. m^* is countably subadditive.

54) $m^*(A_n) = \inf \left\{ \sum_{m=1}^{\infty} l(I_m); A_n \subset \bigcup_{m=1}^{\infty} I_m \right\}$, where
 $\{I_m\}_{m=1}^{\infty}$ is an open covering of A_n

55) For any $\phi \neq A \subseteq \mathbb{R}$, & any $\epsilon > 0$, there
 is an open set "U" such that $A \subset U$
 & $m^*(U) < m^*(A) + \epsilon$

56) $m^*([a, b]) = b - a$ closed interval

57) $m^*(a, b) = b - a$ open interval

58) $b - a \leq m^*(I) \leq b - a$ semi-open interval

59) if $E = \{p \in \mathbb{Q} : 0 < p < 1\}$ & $\{I_n\}_{n=1}^K$ is the finite seq. of open intervals such that $A \subset \bigcup_{n=1}^K I_n$, then $\sum_{n=1}^K l(I_n) > 1$.

60) A $\phi \neq \emptyset E \subset \mathbb{R}$ is said to be Lebesgue measurable or simply measurable if $\forall A \subset \mathbb{R}$.

$$61) m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$$

62) A set E is measurable if & only if its complement (E^c) is measurable.

63) If $m^*(E) = 0$, then $E \subset \mathbb{R}$ is measurable.

64) For any $\phi \neq A \subset \mathbb{R}$ & $\{E_i\}_{i=1}^n$ be a family of pairwise disjoint measurable sets, then

$$m^*[A \cap (\bigcup_{i=1}^n E_i)] = \sum_{i=1}^n m^*(A \cap E_i) \quad n \in \mathbb{N}$$

65) Truth if $n = K-1$, also implies its truth for $n = K$, $\mathbb{R}$

66) Class $\mathcal{M}$ of all measurable sets is a σ -algebra.

$$67) \{E_i\}_{i=1}^{\infty} \Rightarrow \bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$$

68) If F is measurable set &
 $m^*(F \cap G) = 0$ then G is measurable set.

69) A $\phi \neq A \subseteq \mathbb{R}$ is said to be G_δ -set if
it can be written as countable intersection
of open sets.

e.g: $\bigcap_{n=1}^{\infty} (a - \frac{1}{n}, b + \frac{1}{n})$, (a, b) is trivially
 G_δ -set.

70) A $\phi \neq A \subseteq \mathbb{R}$ is said to be F_σ -set if it
can be written as countable union of closed
sets.

e.g

$(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}]$ (a, b) is
also F_σ -set.

71) G_δ -set is measurable being the countable
intersection of measurable sets.

72) F_σ -set is measurable being the countable
intersection of measurable sets (closed).

73) $G'_\delta = F_\sigma$ & $F'_\sigma = G_\delta$

78) Given any $\phi \neq A \subseteq \mathbb{R}$ & $\epsilon > 0$ there is
a G_δ set " G " with $A \subseteq G$ &
 $m^*(A) = m^*(G)$.

79) Interval (a, ∞) is measurable $\forall a \in \mathbb{R}$.

80) (a, ∞) is measurable $\Rightarrow (a, \infty)^c = (-\infty, a]$ is also measurable.

81) $(-\infty, b - \frac{1}{n}]$ is also measurable $\forall n \in \mathbb{N}$ As $n \rightarrow \infty$

82) $U = \bigcup_{n=1}^{\infty} \{I_n\}$ each open interval is measurable & countable union of measurable is measurable.

83) $(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - \frac{1}{n}]$ is also measurable, being countable union of measurable set.